

Locating the Prime Carrier: the ζ Prime Signal is the Value-Space Pair Correlation of the Zeros

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Companion to *Spectral Rigidity and Shuffled Spacing Discriminant* — SIP License v1.1

Abstract

The shuffle-knife discriminant established that, among the standard spectral witnesses of the Riemann zeros, the arithmetic prime-resonance is the one carrying ζ -specific structure (FUNCTION, $\sim 11,500\sigma$) while the distributional witnesses read only the universality class (FORM). It located the signal by *elimination*. This note locates it by *construction*: it identifies, positively and to numerical precision, the single two-point object the prime signal lives in. The prime-resonance power $\mathcal{P} = \sum_p |\sum_j e^{-i(\log p)\gamma_j}|^2/N$ is, by expansion, a linear functional of the empirical value-space pair correlation of the zero *positions*; reconstructing $\mathcal{P}$ from that pair correlation alone recovers 100.0% of the signal (exact to machine precision), whereas surrogates that preserve the two-point statistics of the *spacings* or the index-space fluctuations recover only $\sim 7\%$. A fine sweep of the position form factor peaks only at frequencies $\log n$ with n a prime power, with heights tracking the explicit-formula weight $(\Lambda(n)/\sqrt{n})^2$ at Pearson $r = 0.9975$ and vanishing (~ 0.01) at composite n . The separation from the shuffled floor grows from $18\times$ at $N = 200$ to $474\times$ at $N = 10^4$. The operational conclusion is that “the primes are not a two-point statistic” — a claim we had entertained internally from spacing-domain surrogate tests — is false and is retracted here: the primes are exactly the two-point pair correlation of the positions, i.e. the Montgomery form factor read through the Riemann–von Mangoldt explicit formula. The word “two-point” is simply ambiguous between the positions and the spacings; the carrier is the former, and the surrogate collapse measures the destruction of that object, not the presence of any higher-order one.

Publication status (24 September 2026). This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

1 The ambiguity that motivated the test

The prime-resonance witness of the shuffle-knife study is a scalar built from the actual zero heights,

$$\mathcal{P} = \sum_p P(\log p), \quad P(f) = \frac{1}{N} \left| \sum_{j=1}^N e^{-if\gamma_j} \right|^2, \quad (1)$$

summed here over $p \in \{2, 3, 5, 7, 11, 13\}$, with $\gamma_1 < \dots < \gamma_N$ the ordinates of the nontrivial zeros. The knife showed that $\mathcal{P}$ collapses under a marginal-matched permutation of the spacings $s_i = \gamma_{i+1} - \gamma_i$: the signal is destroyed when the *order* of the spacings is destroyed. A natural

but unguarded reading of that collapse — recorded in a working synthesis during the elimination campaign (see `docs/Elimination_Ledger.md`) — is that the prime signal is inaccessible to any two-point statistic and must reside in higher-order phase coherence, so that recovering it would require a genuinely new, higher-order instrument (a “converse machine”: an instrument built to detect exactly the structure a destructive test showed was being destroyed).

That reading conflates two different two-point objects. Expanding (1),

$$P(f) = \frac{1}{N} \sum_{j,l} e^{-if(\gamma_j - \gamma_l)} = \int e^{-ifs} \rho_2(s) ds, \quad \rho_2(s) = \frac{1}{N} \sum_{j,l} \delta(s - (\gamma_j - \gamma_l)), \quad (2)$$

so $P(f)$ is by construction the Fourier transform of ρ_2 , the empirical *value-space pair correlation of the positions*. It is a two-point statistic — of the positions. The shuffle knife destroys a *different* two-point object, the correlation of the spacing sequence, which does not determine ρ_2 because the map spacings $\mapsto$ positions is a cumulative sum. Whether the primes are “two-point” therefore has no answer until one says two-point *of what*. This note settles it by direct measurement.

Proposition 1 (Pair-correlation sufficiency). *$\mathcal{P}$ is a linear functional of ρ_2 alone; hence any transformation of the point set that preserves the value-space pair correlation of the positions preserves $\mathcal{P}$ exactly.*

This is immediate from (2); the content of the paper is the measurement of *which* surrogate operations preserve ρ_2 and which annihilate it, and the identification of the peak structure with the explicit formula.

2 Method

All measurements use the functional (1) — the complex-exponential power form of the shuffle-knife suite’s prime-resonance witness (FF06e uses the cosine form $\langle \cos(\gamma \log p) \rangle^2$ over $p \leq 29$; (1) is its coherent-power generalisation, and the prime set here is $p \leq 13$) — with the canonical seed 20260423 and the first N ordinates of the Riemann zeros. We evaluate the following, each a distinct answer to “two-point of what?”:

Baseline. $\mathcal{P}$ on the ordered zeros.

Spacing floor. $\mathcal{P}$ after a random permutation of $\{s_i\}$ (one-point spacing distribution only). Averaged over 300 permutations.

Gap two-point (C). $\mathcal{P}$ after phase randomisation of the spacing series, which *preserves the spacing autocovariance* (the two-point statistic of the gaps) and destroys all higher-order gap structure.

Index two-point (D). $\mathcal{P}$ after phase randomisation of the position fluctuation about a smooth degree-5 counting trend, preserving the index-space autocovariance of the positions.

Value-space pair correlation (E). $\mathcal{P}$ recomputed from ρ_2 via (2) (exact, and a 4000-bin histogram variant).

Phase randomisation preserves a power spectrum, hence exactly a two-point autocovariance, while destroying phase relations; it is the standard surrogate for isolating what a two-point statistic can and cannot carry. As calibration we require (i) the ρ_2 reconstruction to reproduce the baseline to machine precision, and (ii) a Poisson point process of matched mean density to show no prime peak above the floor.

3 Results

Which two-point object carries the signal (N=200, 300 trials).

object preserved	$\mathcal{P}$	% baseline
ordered baseline	51.99	100.0%
spacing shuffle (one-point)	2.77	5.3%
gap two-point (C, spacing autocovariance)	3.77	7.3%
index two-point (D, position-fluctuation spectrum)	3.73	7.2%
value-space pair correlation (E, exact)	51.99	100.0%
value-space pair correlation (E, 4000-bin)	51.66	99.4%
Poisson null (calibration)	5.87	11.3%

The pair-correlation reconstruction reproduces the baseline to $< 10^{-9}$ (calibration passed); the coarse 4000-bin histogram still recovers 99.4%. The two spacing/index two-point surrogates recover essentially nothing. *The primes are the value-space pair correlation of the positions; they are absent from the gap and index two-point objects.*

The peaks are the explicit formula (N=4000). A fine sweep of the position form factor $K(f) = |\sum_j e^{-if\gamma_j}|^2/N$ over $f \in (0.5, \log 40)$ shows peaks *only* at $f = \log n$ with n a prime power, and silence at composite n . Measured peak heights track the Riemann-von Mangoldt weight $\Lambda(n)/\sqrt{n}$, where Λ is the von Mangoldt function ($\log p$ at prime powers $n = p^k$, else 0):

n	type	measured height	$\Lambda(n)/\sqrt{n}$
5	prime	66.0	0.720
7	prime	69.3	0.736
11	prime	64.8	0.723
13	prime	61.3	0.711
4	prime power 2^2	14.1	0.347
8	prime power 2^3	6.9	0.245
9	prime power 3^2	15.4	0.366
16	prime power 2^4	3.7	0.173
32	prime power 2^5	1.6	0.123
6, 10, 12, 15	composite	~ 0.00	0

Mean height at prime-power frequencies is 36.6 against 0.01 at composite frequencies. The correlation between measured prime-power peak height and $(\Lambda(n)/\sqrt{n})^2$ is

$$\text{Pearson } r = 0.9975.$$

Heights fall with n exactly as $1/\sqrt{p^k}$ (so 4, 8, 9, 16, 32 shrink monotonically) and vanish where $\Lambda = 0$ (composites). This is the explicit-formula coupling between zeros and prime powers, measured directly on the form factor; it also disposes of the collateral internal conjecture that the two-point form factor $K(\tau)$ would fail to show the primes.

Robustness in N .

N	baseline $\mathcal{P}$	shuffle floor	separation
200	51.99	2.83	18.4×
500	88.68	2.13	41.7×
1000	137.53	1.87	73.6×
2000	216.40	1.75	123.4×
5000	407.69	1.53	266.9×
10000	672.90	1.42	474.2×

The value-space reconstruction equals the baseline at every N (Proposition 1); the separation from the floor (each floor averaged over 200 spacing permutations) sharpens with N , consistent with the shuffle-knife amplification result. The floor is a small, mildly noisy quantity; the separations are reproducible from `scripts/prime_carrier_reproduce.py` at the canonical seed but should be read as $\mathcal{O}(10)$ -to- $\mathcal{O}(10^2)$ growth, not to the last digit.

Coherence and mechanism. The raw peak $|\sum_j e^{-i(\log 2)\gamma_j}|^2$ grows faster than N ($|\cdot|^2/N$: $8.1 \rightarrow 50.5$ across $N = 500 \rightarrow 8000$; $|\cdot|^2/N^2$: $0.016 \rightarrow 0.006$), i.e. the peak is a coherent, non-self-averaging arithmetic sum. And the gap two-point surrogate is *faithful*: the spacing autocovariance at lags 1, 2, 3 is $(-0.009, 0.061, 0.067)$ on the object and $(-0.009, 0.061, 0.067)$ on the surrogate, yet $\mathcal{P}$ falls from 216.4 to 1.5. A surrogate that provably preserves the gap two-point object still loses the primes — the carrier is the position pair correlation, not the gap correlation.

4 What this establishes, and the honest edges

Establishes. The ζ prime-resonance signal is, positively and to machine precision, the value-space pair correlation of the zero *positions* — equivalently the single-realisation spectral form factor — whose peaks are the prime powers with explicit-formula weights ($r = 0.9975$). The prior internal reading “no two-point statistic carries the primes; a higher-order machine is required” is **retracted**: it measured the wrong two-point object (the spacings), and the surrogate collapse it observed is the destruction of ρ_2 , not the signature of higher-order structure. The “converse machine” is therefore not new: it is the pair correlation / explicit formula already central to the Montgomery–Odlyzko picture.

Nuance, not overclaimed. The signal lives in the *single-realisation* form factor of the actual zeros. The ensemble-averaged pair correlation tends to the smooth GUE law; the prime peaks are the arithmetic, non-self-averaging part (the off-diagonal/diagonal decomposition of the form factor). “Two-point carries the primes” is a statement about the empirical form factor of the real zeros, not about the universal GUE R_2 . This is consistent with the shuffle knife: the distributional witnesses are FORM because they see only the universal part; the arithmetic witness is FUNCTION because it sees the non-universal peaks that *are* ρ_2 ’s deviation from GUE.

Where the method stops. The identification is spectral and requires enough resolved zeros for the peaks to stand above the floor. For an L -function with only a handful of reachable zeros (e.g. the elliptic curve 11a1, with ~ 10 ordinates in the public tables) the form factor cannot resolve its arithmetic peaks at reachable height. That data scarcity is exactly the measurability wall the signal-to-noise criterion names elsewhere in this programme: the machine is known and universal, but the spectral data must exist to run it. We mark this boundary so the positive result is not over-read as a claim about objects whose spectra we cannot yet reach.

Companion to *Spectral Rigidity and Shuffled Spacing Discriminant* and to *Riemann Spectral Critical Line*. All measurements use the real first N Riemann zeros and the canonical seed 20260423; the functional (1) is the coherent-power generalisation of the shuffle-knife prime-resonance witness (cosine form, $p \leq 29$, in FF06e; complex-exponential form, $p \leq 13$, here), and the reconstruction/sweep are reproducible standalone. Nothing here proves the Riemann Hypothesis or constructs a Hilbert–Pólya operator; “carrier” means “the two-point object the prime-resonance functional is a functional of,” identified and calibrated. SIP License v1.1.

Amendment: scope and evidence, 24 September 2026

The strong reported correlation supports the stated finite position-based observable and explicit-formula comparison. It does not uniquely identify a microscopic carrier or exclude every arithmetic contribution to other observables. Truncation, smoothing, normalization and the sampled interval remain part of the experiment. Extending the result to an infinite zero sum requires convergence control beyond correlation. The new physical carrier audit concerns a separate field-theoretic identification problem and supplies no missing RH premise.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s [papers/updates/](#) directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.